Birla Institute of Technology & Science
Pilani Campus

Work-Integrated Learning Programmes Division

KDE Connect Web Client

Capstone Project (S1-25_BSDCHZC499T)

Vaisakh KM

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STUDENT'S EMPLOYING ORGANIZATION & LOCATION	**: HCLTech, Chennai**
MENTOR'S NAME	**: Venkata Santhosh Naga Sai**
MENTOR'S EMPLOYING ORGANIZATION & LOCATION	**: HCLTech, Bangalore**
PROPOSED PROJECT TITLE	**: KDE Connect Web Client**

CAPSTONE PROJECT OUTLINE EVALUATION

(Please put a tick (✓) mark in the appropriate box)

EC No.	Component	Excellent	Good	Fair	Poor
1.	Capstone Project Outline	✓			

	Mentor	Additional Examiner
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BITS ZC499T Capstone Project Outline

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CAPSTONE PROJECT TITLE : **KDE Connect Web Client**

Capstone Project Outline

Student Name : Vaisakh K M
Registration Number : 202117B3543
Course Number : Capstone Project (Cohort 8)
Course Title : B.Tech Capstone Project
Capstone Project Title : KDE Connect WebApp – Bringing cross-device integration to the browser with a lightweight, embedded server architecture
Broad Academic Area of Work : Computer Science & Engineering (Systems Software / Cross-Platform Integration / Web Technologies)

Background

KDE Connect is a widely-used open-source application that enables seamless workflow integration across devices, offering features such as clipboard sharing, notification mirroring, remote input control, file sharing, media control, and more. It currently supports Qt-based desktop clients (Linux/Windows/macOS), Android, and iOS platforms.

Despite its popularity, there is no official web client, which significantly limits accessibility for users in restricted environments (corporate laptops, public computers, Chromebooks, or secondary devices) where installing native applications is not feasible or permitted. This creates a major usability gap: users must install KDE Connect on every device they wish to integrate, reducing convenience and adoption in multi-device workflows.

The proposed project addresses this gap by developing a fully functional web client for KDE Connect, requiring installation on only one device (typically the phone or primary computer). The solution runs a lightweight embedded server within the native KDE Connect app and exposes all core functionalities securely to any modern browser via a responsive web interface.

Objectives

1. Port the KDE Connect protocol stack to work reliably in web environments, including implementation of custom TLS encryption suitable for browser constraints.

2. Develop a lightweight Kotlin-based HTTP/WebSocket server that can run embedded within the existing KDE Connect Android (and optionally desktop) applications with minimal resource overhead.

3. Build a responsive, feature-rich web client using modern browser APIs (File System Access API, WebSocket, Clipboard API, Notifications API, etc.) to deliver core KDE Connect functionality directly in the browser.

4. Design the server as a modular plugin for existing KDE Connect installations, enabling seamless integration without requiring users to replace or modify their current setup.

5. Ensure secure communication using self-signed certificates with QR-code-based trust establishment, while maintaining high performance even on resource-constrained mobile devices.

Scope of Work (to be done independently by the student)

- In-depth analysis and reverse-engineering of KDE Connect network protocols and packet structures.
- Design and implementation of a custom TLS layer compatible with both native KDE Connect and browser security requirements.
- Development of a complete embedded Kotlin server (using Ktor/nanohttpd) with WebSocket and REST endpoints that translate native KDE Connect packets to web-friendly formats.
- Creation of a full-stack web client in Svelte (or equivalent modern framework) with responsive UI, real-time updates, file browsing, clipboard sync, notification handling, remote input, and media controls.
- Implementation of secure pairing via QR code with self-signed certificate trust propagation.
- Optimization for low memory/CPU usage on Android devices (connection pooling, adaptive throttling, foreground-service awareness).
- Cross-browser testing (Chrome, Firefox, Safari) and cross-platform testing (Android + Windows/Linux/macOS hosts).
- Security audit and penetration testing of the implementation.
- Documentation, packaging as an installable plugin for KDE Connect Android, and preparation for community contribution.

Plan of Work (15-Week Semester Timeline)

Weeks 1–3: Research & Design

- Detailed protocol analysis of KDE Connect (network packets, encryption, pairing)
- Architecture design and API specification
- Feasibility studies for browser APIs and Android restrictions
- Finalize tech stack and plugin integration strategy

Weeks 4–6: Server Development

- Implement lightweight Kotlin server (Ktor/nanohttpd)
- TLS implementation with self-signed certificates
- Plugin architecture for KDE Connect Android app
- Basic WebSocket + REST endpoints

Weeks 7–9: Protocol Porting

- Full porting of KDE Connect identity, pairing, and capability exchange packets
- Implementation of core plugins (clipboard, notifications, battery, telephony, filesystem)
- End-to-end encrypted channel testing between server and native clients

Weeks 10–12: Web Client Development

- Responsive UI in Svelte
- Integration of FileSystemAccess API, Clipboard API, Notification API, WebSocket
- Implementation of clipboard sync, notification mirroring, file browser, remote input, find-my-phone, media controls, etc.
- QR-code pairing and certificate trust flow

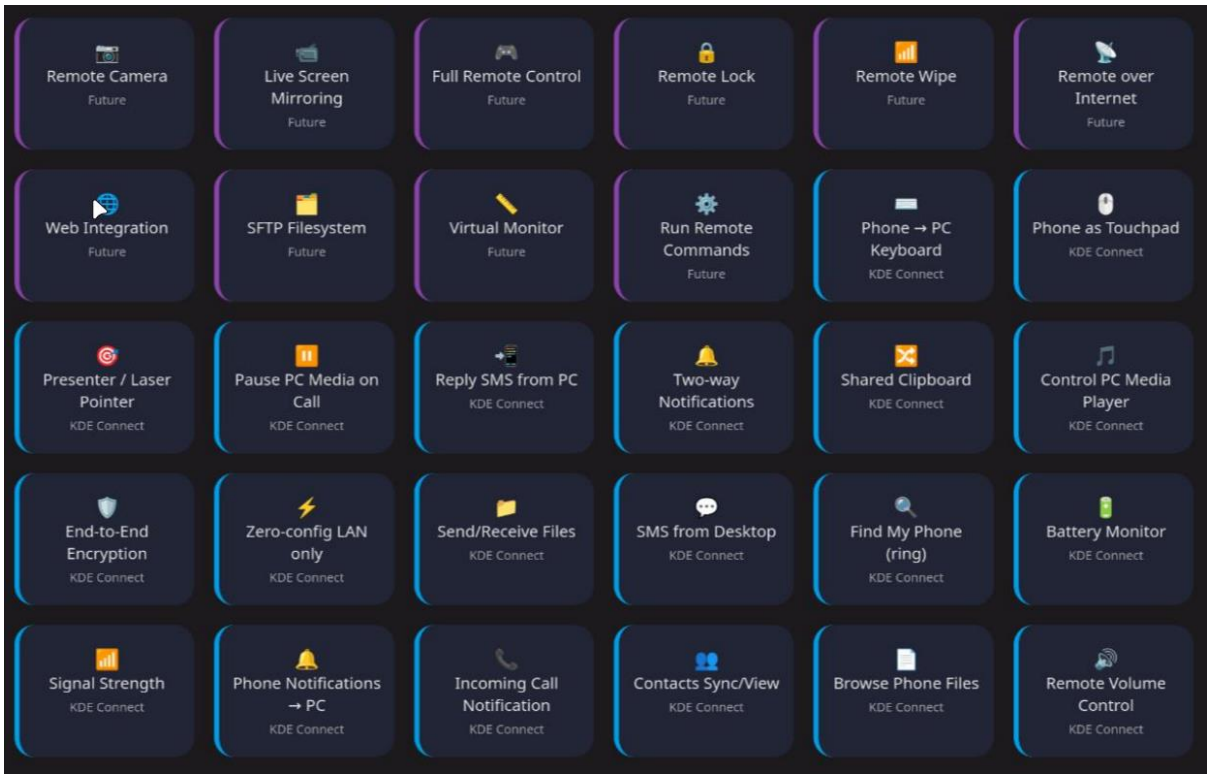
Weeks 13–14: Testing & Optimization

- Cross-platform and cross-browser testing
- Performance profiling and optimization (memory, battery, CPU)
- Security audit and bug fixing
- Usability testing

Week 15: Documentation & Release

- Complete user and developer documentation
- Packaging as APK plugin and web app bundle
- Prepare GitHub repository and community rollout (KDE Invent submission)

Planned Features



Literature References / Resources

1. KDE Connect Official Repository – https://invent.kde.org/network/kdeconnect-kde
2. KDE Connect Android Repository – https://invent.kde.org/network/kdeconnect-android
3. KDE Connect Protocol Documentation – https://community.kde.org/KDEConnect#Protocol
4. Ktor Framework Documentation – https://ktor.io
5. Svelte Documentation – https://svelte.dev/docs
6. File System Access API – MDN Web Docs
7. WebSocket API – MDN Web Docs
8. Android Foreground Service Guidelines (Android 12+)
9. BoringSSL / OpenSSL self-signed certificate handling references
10. Nanohttpd lightweight server library

Signature of Student	**Signature of Mentor**	**Signature of Additional Examiner**
Name: Vaisakh KM	**Name: Venkata Santhosh Naga Sai**	**Name: Shershad MP**